University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 25
Program: B.Sc. in CSE
2nd Year 2nd Semester

Course Title: Probability and Statistics | Course Code: MTH 203
Time: 1 hour | Credit Hour: 3.00 | Full Marks: 20

Answer all the questions

QUESTION 1 [10 MARKS]

1.(a) Define geometric and harmonic mean. Determine the median, standard deviation, and coefficient of variation from the following data: [05]

Marks	20-29	30-39	40-49	50-59	60-69	70-79
No. of students	2	20	35	20	10	10

(b) Using the frequency distribution of question 1(a), draw a histogram and an ogive curve. Prove that, for a series of n observations, the standard deviation of first n natural number is [05]

$$\sqrt{\frac{n^2-1}{12}}.$$

QUESTION 2 [10 MARKS]

2.(a) Define conditional probability and Bayes' theorem. In a certain town, males and females each form 50% of the population. It is known that 15% of the males and 8% of the females are unemployed. A research student studying the employment situation selects an unemployed person at random. What is the probability that the person so selected is male? [05]

(b) A can hit a target 3 times in 5 shots, B 2 times in 5 shots, and C 3 times in 4 shots. All of them fire 1 shot each simultaneously at the target. What is the probability that (i) 2 shots hit (ii) at least 2 shots hit? [05]

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Program: B.Sc. in CSE
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Course Title: Data Structures & Algorithms II — Course Code: CSE 207
Time: 1 hour. — Credit Hour: 3.00 — Full Marks: 20

There are Three Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [3 MARKS]

a. Explain the precondition and all possible cases of Ternary search algorithm. [3]

QUESTION 2 [7 MARKS]

a. You're coordinating a helicopter supply drop to a remote village after a disaster. The helicopter can carry 43 kg of supplies. You have 3 types of supply boxes mentioned in **Table 1**. You can take fractions of any box or whole and your goal is to maximize the total importance value loaded. [5]

Table 1. Supply box Information

Supply Box	Weight (kg)	Importance value
A	20	240
B	30	270
C	15	150

Using the fractional knapsack approach, calculate the total maximum importance value you can carry and justify why the fractional knapsack method is more preferable than the 0/1 knapsack method.

b. You've joined a team that's developing a nature-themed mobile game for children. In one of the mini-games, a character - a curious rabbit discovers a magical garden. The garden has a strange planting pattern: each day, the number of new flowers blooming follows a unique rule. [2]

On Day 0, 0 flowers bloom.

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On Day 1, 1 flower blooms.

From Day 2 onward, the number of flowers blooming is the sum of the number of flowers from the previous two days.

Your task is to apply an algorithm that helps to visualize this pattern.

QUESTION 3 [2 MARKS]

a. Analyze the recurrence relation of the algorithm used in Question 2(b) and check whether it is possible to calculate the time complexity of that algorithm using Master theorem or if not, give proper explanation. [2]

QUESTION 4 [8 MARKS]

a. You are helping a local library build a digital cataloging system. As part of a data-cleaning task, you are given a list of 8 book titles, but instead of sorting by the full names, your manager asks you to perform a special task: [5]

"Take the first letter of the first book, the second letter of the second book, the third letter of the third book, and so on... all the way to the eighth letter of the eighth book. Once you have those 8 letters, sort them alphabetically using Quicksort to arrange the books.

Book list : "Go", "bear" , "coff" ,"appa", "music","launch" ,"baabbyy", "bridepad" .

b. Once you get that sorted sequence of 8 letters from Question 4(a), find the Longest Common Subsequence (LCS) between: [3]

The sorted sequence of letters you extracted and a reference word or sequence provided by your manager (assume the manager gives you a secret code word like "bracelet")

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Mid Semester Examination, Spring 2025
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Course Title: Engineering Economics — Course Code: ECN 201
Time: 1 hour. — Credit Hour: 3.00 — Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. Explain the factors affecting demand. [5]

b. $Q = 100-2P$ [5]

$Q= 50+2P$

Calculate equilibrium price and quantity.

QUESTION 2 [10 MARKS]

a. Describe different types of elasticity of demand. [5]

b. Calculate elasticity and give interpretation. [5]

Q	P
10	5
15	10
25	12
30	15
40	20

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Mid Semester Examination, Spring 25
Program: B.Sc. in CSE
2nd year 2nd Semester

Course Title: Digital Logic Design | Course Code: CSE 209
Time: 1 hour. | Credit Hour: 3.00 | Full Marks: 20

There are **Three** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [7 MARKS]

a. Discuss the universality of NAND gate. [2]

b. Implement the following Boolean function with only NAND gate [2]
$y = A`C + AB` + BC`$

c. Simplify the following function using K-map. [3]
$F(A, B, C, D) = \Sigma(0, 1, 2, 3, 7, 8, 10, 11, 13, 14, 15)$

QUESTION 2 [7 MARKS]

a. Design a 4-bit parallel adder/subtractor using IC # 7483 and other logic gates if necessary. Briefly describe its operation. [3]

b. Design a MOD-16 up/down counter using J-K flipflops and other logic gates if necessary. Briefly describe its operation. [4]

QUESTION 3 [6 MARKS]

a. Draw the internal circuit diagram of J-K flipflop and briefly describe its operation considering the truth-table. [3]

b. Draw the internal circuit diagram of D flipflop and briefly describe its operation. [2]

c. Convert a J-K flipflop into a D flipflop. [1]

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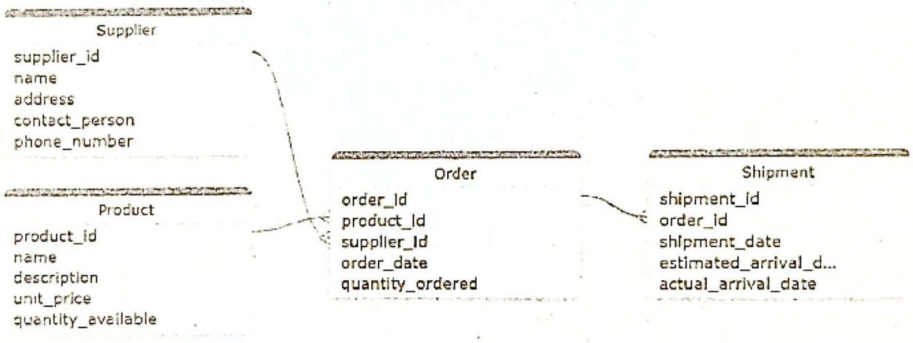
University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 25
Program: B.Sc. in CSE
2nd year 2nd Semester

Course Title: Database Systems | Course Code: CSE 211
Time: 1 hour. | Credit Hour: 3.00 | Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. The following relational schema forms a student database which is implemented in a relational database: [6]



Construct the Relational Algebra for the following queries:

i) List the supplier's name who ordered 100 items.

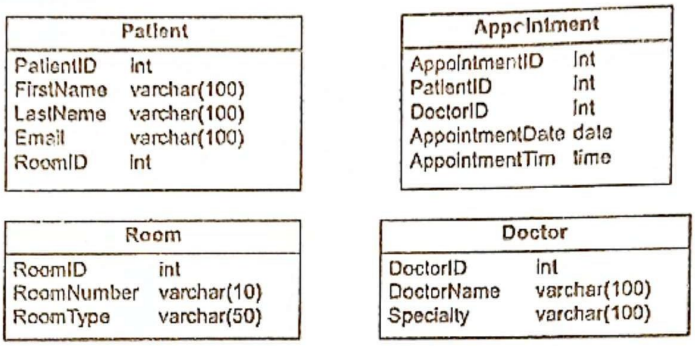
ii) List the name of the products where unit price range of $50 - $60 and available quantity should be more than 20.

iii) List the shipment id where shipment date is '12-02-2025' or estimated arrival date is '28-02-2025' using Set Operation.

b. Explain briefly 2-Tier and 3-Tier architecture of a database system with appropriate examples. [2]

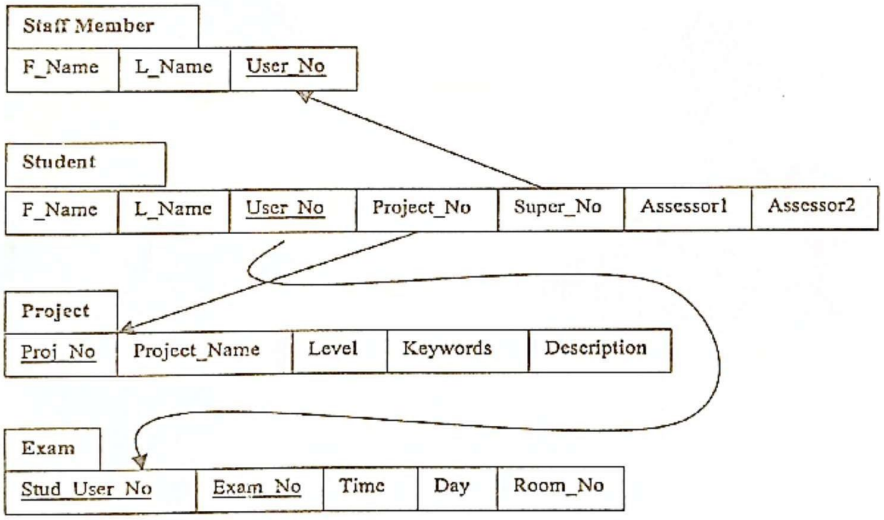
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c. Identify the primary keys, foreign keys in the following scenario: [2]



QUESTION 2 [10 MARKS]

a. Construct the DDL (create) for the tables below with the necessary data-types and constraints. [4]



b. Construct the DML for the following queries using the schema provided in Question 2(a). [6]

i) Insert one record in the student table.

ii) What are Exam_No and Room_No for Stud_User_No 202?

iii) Remove 101 User_No's record from Staff_Member table.

iv) Update Assessor1 as 'Rafi" for User_No 445.

v) List out all F_Name and L_Name of Staff_Member

vi) Show the Exam record of Sunday.

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